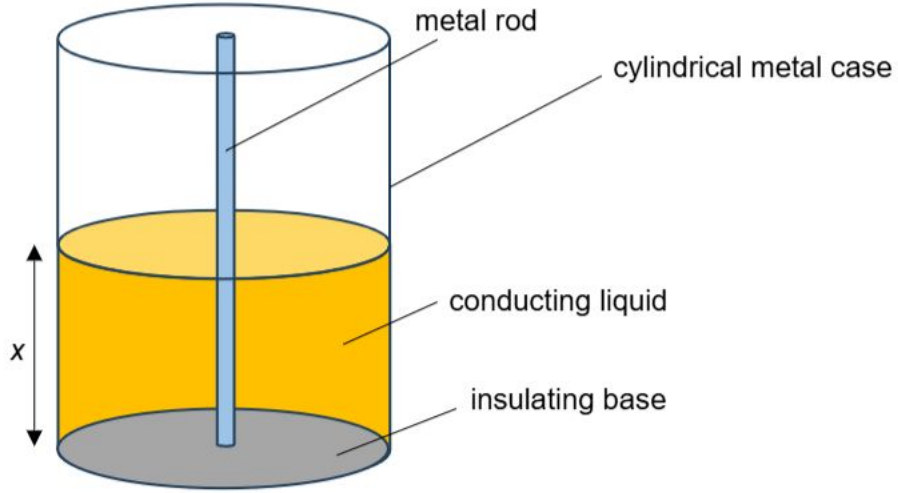
21 A conducting liquid fills a cylindrical metal case to a depth x as shown below. The case has an insulating base.



The resistance between the case and the metal rod will be

A proportional to x^2

B proportional to x

C inversely proportional to x

D inversely proportional to x^2

Ans: **C**

$$R = \rho\frac{l}{A} = \rho\frac{r}{2\pi r_{average}x} \rightarrow R \propto \frac{1}{x}$$